

Saeed Odak

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Last updated: May 29, 2026

Education

May 2025–Present	Postdoctoral Researcher, Computational Geometry Group, Aalto University, Finland
Jan. 2021–May 2025	Ph.D. Computer Science, University of Ottawa, Canada
Sep. 2014–Aug. 2019	B.Sc. Mathematics, K. N. Toosi University of Technology, Iran
Sep. 2014–Aug. 2019	B.Sc. Electrical Engineering and Telecommunication, K. N. Toosi University of Technology, Iran
Sep. 2010–Aug. 2014	High School Diploma (Math and Physics), NODET, Gorgan, Iran

Academic Awards

Fall 2023	PhD Mobility Program in France (MDF) Research Scholarship, University of Bordeaux
2021–2025	International Doctorate Scholarship, University of Ottawa
2021–2024	Ph.D. Admission Scholarship, University of Ottawa
2019	Ranked 2nd among all B.Sc. students of Mathematics, K. N. Toosi University of Technology
2019	Ranked 7th in National Entrance Exam for M.Sc. in Computer Science, Tehran, Iran
2018	Bronze Medal, International Mathematics Competition (IMC), Blagoevgrad, Bulgaria
2018	Silver Medal, Iranian Mathematics Competition, Behshahr, Iran
2017	Bronze Medal, Iranian Mathematics Competition, Shahr-e Kord, Iran
2019	Silver Medal, ACM-ICPC Asia Tehran Regional Contest (team), Sharif University
2018	7th place, ACM-ICPC Asia Tehran Regional Contest (team), Sharif University
2021	29th place, IEEE-Xtreme 15.0 International Programming Contest (team)

2018 27th place, IEEE-Xtreme 12.0 International Programming Contest (team)

Relevant Work Experience

Postdoctoral Researcher May 2025–Present
Aalto University, Espoo, Finland
 Research on computational geometry, hyperbolic geometry, fine-grained geometric lower bounds and algorithmic graph theory.

Research Assistant Sept.–Nov. 2023
LaBRI, University of Bordeaux, France
 Visiting research on Euclidean freeze tag problem and universal induced subgraphs.

Research Assistant Jan. 2021–May 2025
Algorithms, Graphs, and Geometry Lab (AGGLAB), Carleton University, Canada
 Research on algorithmic and structural graph theory, proximity data structures, graph drawing, computational geometry.

Undergraduate Researcher May –Aug. 2019
K. N. Toosi University of Technology, Iran
 B.Sc. thesis on Routing and Wavelength Assignment (RWA) in Optical Networks, supervised by Lotfollah Beygi.

Current Research Interests

Structural and algorithmic graph theory	Combinatorial, structural, and algorithmic problems on graphs, including product structure, graph drawing, planarity, and graph colouring [4,5,6,7,12,13,16]
Computational geometry	Algorithmic and combinatorial problems in discrete and computational geometry, including spanners, longest paths, art gallery problems, and freeze-tag [3,8,9,10,11,14,15]
Hyperbolic geometry	Algorithms for fundamental problems in hyperbolic geometry, including clustering, TSP and Steiner tree [1]
Hardness results and fine-grained complexity	Hardness results and fine-grained complexity for geometric and graph problems [3]
Data structures	Design and analysis of geometric proximity structures [11].

Professional Duties

Conference Reviewing	ISAAC 2022, SoCG (2025, 2026), CCCG (2022, 2023, 2024, 2025, 2026), WADS (2023, 2025), SWAT (2024, 2026), CALDAM (2024), SoSA (2026), ESA (2026), WG (2026), GD (2024, 2026), LAGOS (2025), ISAAC (2022)
Journal Reviewing	ACM Transactions on Algorithms, Computational Geometry: Theory and Applications; Computing in Geometry and Topology

Teaching and Supervision

Student Supervision

- Marc Vicuna (M.Sc.) School of Computer Science, Carleton University
May 2022 – April 2024 Thesis: *Efficient Computation of Interesting Paths*.
- Satu Koitto (B.Sc.) Department of Computer Science, Aalto University
May 2024– Aug. 2024 Thesis: *Linear and Centered Chromatic Number of Graphs*.

Contract Instructor

Jan.–Apr. 2023

Carleton University – Discrete Structures I (COMP 1805)

Teaching Assistant

Jan. 2021–Dec. 2024

University of Ottawa – Design and Analysis of Algorithms; Data Structures and Algorithms; Discrete Structures; Introduction to Computing I & II; Introduction to Formal Languages.

Teaching Assistant

Sep. 2016–Aug. 2019

K. N. Toosi University of Technology – Algorithm Design; Data Structures; Calculus I & II.

Volunteer Teaching

Sep. 2016–Dec. 2020

K. N. Toosi University of Technology – Coaching ICPC team (programming competition: Algorithms and Data Structures)

Publications

Submitted Papers

- [1] Manuel Cásares, Sándor Kisfaludi-Bak, and **Saeed Odak**. Shifting is Optimal under Gap-ETH: A Lower Bound Framework for Geometric Approximation Schemes. *Submitted to European Symposium on Algorithms (ESA 2026)*.
- [2] Ahmad Biniiaz, Sujoy Bhore, Kevin Buchin, Antonia Kalb, Anil Maheshwari, **Saeed Odak**, Carolin Rehs, Michiel Smid. Computing oriented spanners and their dilation. In *Submitted to European Symposium on Algorithms (ESA 2026)*.
- [3] **Saeed Odak**, Jonathan Rollin, and Torben Scheele. On 2-Layer k -Matching-Planar Graphs. Submitted to *Graph Drawing (GD 2026)*.
- [4] Jean-Lou De Carufel, Anil Maheshwari, **Saeed Odak**, Bodhayan Roy, Michiel Smid, and Marc Vicuna. Deciding if a DAG is interesting is hard. Submitted to *Computing in Geometry and Topology (CGT)*. CoRR, abs/2503.13398, 2025.
- [5] Ahmad Biniiaz, Jean-Lou De Carufel, Anil Maheshwari, **Saeed Odak**, André van Renssen, Michiel Smid, and Sampson Wong. Euclidean TSP with Mandatory Returns to a River Line. *Submitted to 51st International Symposium on Mathematical Foundations of Computer Science (MFCS 2026)*.

Papers Accepted in Refereed Journals

- [6] Greg Aloupis, Ahmad Biniiaz, Prosenjit Bose, Jean-Lou De Carufel, David Eppstein, Anil Maheshwari, **Saeed Odak**, Michiel Smid, Csaba D. Tóth, and Pavel Valtr. Noncrossing longest paths and cycles. *Graphs and Combinatorics*, 41(6):122, 2025. Preliminary version appeared at GD 2024.

- [7] Michael A. Bekos, Prosenjit Bose, Aaron Büngener, Vida Dujmović, Michael Hoffmann, Michael Kaufmann, Pat Morin, **Saeed Odak**, and Alexandra Weinberger. On k -planar graphs without short cycles. *Journal of Graph Algorithms and Applications*, 29(3):1–22, 2025. Preliminary version appeared at GD 2024.
- [8] Ahmad Biniaz, Prosenjit Bose, Jean-Lou De Carufel, Anil Maheshwari, Babak Miraftab, **Saeed Odak**, Michiel Smid, Shakhar Smorodinsky, and Yelena Yuditsky. On separating path and tree systems in graphs. *Discrete Mathematics & Theoretical Computer Science*, 27(2), 2025.

Papers Accepted at Refereed Conferences

- [9] Prosenjit Bose, Vida Dujmović, Hussein Houdrouge, Pat Morin, and **Saeed Odak**. Connected dominating sets in triangulations. In *The 53rd International Colloquium on Automata, Languages, and Programming (ICALP 2026)*.
- [10] Sándor Kisfaludi-Bak, **Saeed Odak**, Satyam Singh, and Geert van Wordragen. Gap-ETH-Tight Algorithms for Hyperbolic TSP and Steiner Tree. In *42nd International Symposium on Computational Geometry (SoCG 2026)*, LIPIcs, pages 64:1–64:17. Schloss Dagstuhl, 2026.
- [11] Ahmad Biniaz, Prosenjit Bose, Chaeyoon Chung, Jean-Lou De Carufel, John Iacono, Anil Maheshwari, **Saeed Odak**, Michiel Smid, and Csaba D. Tóth. Tight bounds on the number of closest pairs in vertical slabs. In *19th Algorithms and Data Structures Symposium (WADS 2025)*, LIPIcs, pages 8:1–8:14. Schloss Dagstuhl, 2025.
- [12] Ahmad Biniaz, Anil Maheshwari, Magnus Christian Ring Merrild, Joseph S. B. Mitchell, **Saeed Odak**, Valentin Polishchuk, Eliot W. Robson, Casper Moldrup Rysgaard, Jens Kristian Refsgaard Schou, Thomas C. Shermer, Jack Spalding-Jamieson, Rolf Svenning, and Da Wei Zheng. Polynomial-time algorithms for contiguous art gallery and related problems. In *41st International Symposium on Computational Geometry (SoCG 2025)*, volume 332 of *LIPIcs*, pages 20:1–20:21. Schloss Dagstuhl, 2025.
- [13] Kevin Buchin, Antonia Kalb, Anil Maheshwari, **Saeed Odak**, Carolin Rehs, Michiel Smid, and Sampson Wong. Computing oriented spanners and their dilation. In *41st International Symposium on Computational Geometry (SoCG 2025)*, volume 332 of *LIPIcs*, pages 27:1–27:17. Schloss Dagstuhl, 2025.
- [14] Nicolas Bonichon, Cyril Gavoille, Nicolas Hanusse, and **Saeed Odak**. Euclidean freeze-tag problem on plane. In *The 36th Canadian Conference on Computational Geometry (CCC 2024)*, 2024.
- [15] Greg Aloupis, Ahmad Biniaz, Prosenjit Bose, Jean-Lou De Carufel, David Eppstein, Anil Maheshwari, **Saeed Odak**, Michiel Smid, Csaba D. Tóth, and Pavel Valtr. Noncrossing longest paths and cycles. In *32nd International Symposium on Graph Drawing and Network Visualization (GD 2024)*, volume 320 of *LIPIcs*, pages 36:1–36:17. Schloss Dagstuhl, 2024.
- [16] Michael A. Bekos, Prosenjit Bose, Aaron Büngener, Vida Dujmović, Michael Hoffmann, Michael Kaufmann, Pat Morin, **Saeed Odak**, and Alexandra Weinberger. On k -planar graphs without short cycles. In *32nd International Symposium on Graph Drawing and Network Visualization (GD 2024)*, volume 320 of *LIPIcs*, pages 27:1–27:17. Schloss Dagstuhl, 2024.
- [17] Prosenjit Bose, Pat Morin, and **Saeed Odak**. An optimal algorithm for product structure in planar graphs. In *18th Scandinavian Symposium and Workshops on Algorithm Theory*

(*SWAT 2022*), volume 227 of *LIPICs*, pages 19:1–19:14. Schloss Dagstuhl, 2022.

Other Contributions

- [18] Ahmad Biniiaz, Anil Maheshwari, Joseph S. B. Mitchell, **Saeed Odak**, Valentin Polishchuk, and Thomas C. Shermer. Contiguous boundary guarding. *CoRR*, [abs/2412.15053](#), 2024.
- [19] Vida Dujmović, Pat Morin, and **Saeed Odak**. Odd colourings of graph products. *CoRR*, [abs/2202.12882](#), 2022.